Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Ali Shakouri	Email	coria-consortium@purdue.edu
Institute Name	Purdue University			
Address	1205 Mitch Daniels Boulevard, West Lafayette, Indiana 47907			

2. Technology Overview

Title of Proposed Technology	**Development of Physical AI-based Nonlinear Error Compensation and Health Monitoring for Ultra-Precision Stages in EUV Inspection Systems**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical · Electronics			
	Medium Category	Small Category	Medium Category	Small Category
	Semiconductor Equipment	*Measurement/Inspection Equipment*		
Concept of Proposed Technology	**Core Concept** The proposed technology is a Physical AI-based precision control solution designed to maximize the performance of ultra-precision stage systems, a critical component of EUV (Extreme Ultraviolet) lithography and inspection equipment. The objective is to overcome the limitations of conventional systems by using Physical AI to model the nonlinear dynamic characteristics of a Dual-Servo system, which integrates a relatively cost-effective Coarse stage with a high-precision, high-response Fine stage. By learning diverse environmental variables and embedding physical laws into neural networks, this technology aims to develop a stage system that achieves nanometer-level control precision resilient to environmental impacts and hardware constraints through real-time compensation of complex interference and nonlinear errors. **Key Functions** 1) Physical AI-based Modeling			

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	● Establishment of a hybrid digital twin that combines the dynamic characteristics of the stage with environmental characteristic models using PINN (Physics-Informed Neural Networks). 2) Integrated Dual-Servo Control ● An intelligent feed-forward control system that learns various system and environmental variables to predict interferences occurring in the Coarse-Fine stage structure in real-time, achieving nanometer-level (<1nm) positioning accuracy. 3) Real-time Prognostics and Health Management (PHM) ● A condition monitoring system that analyzes acceleration and displacement sensor data using AI to proactively detect signs of component wear or failure. **Potential Applications** 1) Next-Generation Semiconductor Manufacturing & Advanced Packaging ● Ultra-precision bonding and inspection for HBM (High Bandwidth Memory). ● High-resolution CD-SEM and non-destructive metrology. 2) Micro-Displays (OLEDoS) & Nano-Lithography ● Ultra-high-resolution deposition for OLEDoS (OLED on Silicon). ● Transfer and bonding processes for Micro-LEDs. 3) Smart Factory PHM & Equipment Operational Efficiency ● Real-time condition monitoring and predictive maintenance. ● Reduction of operational costs and acceleration of domestic component localization.
Rationale for Joint R&D Project	**Critical Technical Barriers** The core technical barrier of this project lies in the difficulty of accurately modeling and compensating for the strongly nonlinear, time-varying dynamics and environmental sensitivity of Dual-Servo (Coarse–Fine) ultra-precision stage systems used in EUV inspection and process equipment using conventional control or data-driven AI approaches. Achieving and maintaining nanometer-level control accuracy under complex environmental disturbances and hardware constraints requires Physical AI-based modeling and control that embed physical laws; however, ensuring stability and reliability when applying such models to real control systems remains highly challenging. Additionally, the limited availability of pre-hardware validation data and the difficulty of predicting long-term performance degradation hinder objective technology maturity validation for global market entry, making these challenges a critical hurdle for this project. **Why Global Collaboration?** Unlike conventional AI, which relies solely on data-driven learning, Physical AI is a highly specialized field that integrates mathematical analysis and physical modeling expertise by using the

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	laws of physics as a guideline for neural network training. Collaborating with a world-renowned overseas research institute in this domain is the most effective strategy to drastically shorten the development cycle and minimize trial and error. Furthermore, establishing a virtual environment system and digital twin within an overseas research institute to verify Physical AI technology enables the algorithm to learn from tens of thousands of scenarios before the actual hardware is even fabricated, thereby maximizing its stability. This global-standard verification process will serve as an objective benchmark for the technological maturity of the EUV inspection systems. It also presents a critical opportunity to secure technical trust from international customers when entering the next-generation semiconductor process market in the future. Ultimately, this international joint R&D will serve as a cornerstone to elevate the driving precision of our equipment to global standards, securing overwhelming performance and cost competitiveness over foreign systems.
Key Words	1. High-Precision Stage 2. Physical AI 3. Digital Twin 4. EUV mask inspection 5. PHM

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (7)

R&D Objectives

R&D Objectives

Development of a Physical AI-based precision control solution to maximize the performance of ultra-precision stage systems, a critical component of EUV lithography and inspection equipment.

- Design and fabrication of a dual-servo based ultra-precision stage system.
- Development of an integrated system for ultra-precision control and PHM (Prognostics and Health Management) based on Physical AI.

R&D Target Specifications

Key Performance Indicators (KPIs)	Target Value	Weight	Measurement and Verification Method
Stage Stroke	350 x 180 mm	10%	Linear encoder measurement
Positioning Accuracy	< 1 nm	30%	Laser interferometer measurement
Dynamic Tracking Error	< 5 nm	30%	Laser interferometer measurement at 100 mm/s constant speed
AI Control Inference Time	< 5 ms	30%	Time required for AI inference

R&D Plan

Year 1: Establishment of Test Environment & Status Assessment

- Construction of the dual-servo testbed
- Collection of baseline nonlinear data
- Establishment of dynamic and environmental variable models

Year 2: Development of Physical AI-based Control Solutions

- Development of control hardware for Physical AI implementation • Model selection
- Optimization of model training for Physical AI time-series error prediction
- Construction of a Digital Twin system based on the trained AI models

Year 3: Verification of Physical AI-based Control and PHM

- Validation of stage control performance
- Field verification of Digital Twin and PHM systems

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute serves as the primary developer of the intelligent control engine, bridging the gap between hardware limitations and required precision through advanced modeling. 1) Physical AI Intelligent Control & Nonlinear Error Compensation: ● Design and develop control models based on Physical AI ● Develop algorithms to compensate for nonlinear errors such as friction, cogging force, and thermal deformation in real-time. 2) High-Precision Digital Twin & Simulation: ● Establish a virtual environment simulation system to model the dual-servo stage's dynamic characteristics. ● Execute large-scale data training within the digital twin to optimize control parameters before physical hardware implementation. 3) Intelligent Prognostics and Health Management (PHM): ● Provide technologies for state inference and predictive maintenance . ● Develop AI-based monitoring logic to detect component wear and predict the remaining useful life of the system.
Expected R&R of Korean SME Partner	The Korean SME partner is responsible for the overall hardware infrastructure, system integration, and the commercialization path within the EUV inspection market. 1) Hardware Design & System Integration: ● Execute the mechanical design and integrated assembly of the entire dual-servo stage system. ● Manage the interface between the Coarse and Fine stages to ensure structural integrity. 2) Coarse Stage Development & Optimization: ● Develop a cost-effective, high-efficiency Coarse stage based on linear motion technology. ● Optimize the hardware to provide stable long-stroke movement as the foundation for the Fine stage. 3) Testbed Provision & Validation: ● Provide a dedicated testbed that simulates a real-world EUV inspection environment. ● Conduct performance validation and mass-production readiness testing to ensure the system meets semiconductor industry standards. 4) Data Collection & Commercialization Strategy: ● Collect raw sensor data (acceleration, displacement) required for training the PHM system. ● Establish a business model for commercializing the technology, focusing on enhancing the competitiveness.

5. Expected Outcomes/Impact

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Expected Outcomes and Impact	**Technological Achievements** 1) Acquisition of Physical AI Control Algorithms ● Beyond the limitations of simple data-driven AI, this project will secure Physical AI control algorithms that reflect the dynamic characteristics of the system in real-time through collaboration with international research institutes. 2) Localization of Nano-scale Precision Technology ● By combining coarse stage with maglev fine stage technology, we will achieve independence in ultra-precision technology, targeting positioning accuracy of less than 1 nm and dynamic tracking error of less than 5 nm, which previously had a high dependence on foreign sources. **Commercialization Potential** 1) Global Price Competitiveness ● By establishing a lineup of high-performance, low-cost inspection equipment, we will secure price competitiveness in the global market and improve profit margins through import substitution effects. 2) Minimization of Process Downtime ● By embedding real-time Health Monitoring (PHM) functions based on Physical AI into our equipment, we can proactively detect fault precursors and perform preventive maintenance, thereby minimizing downtime for our customers' production processes. 3) Integration of High-Value Service Models ● Beyond simple equipment sales, we will integrate high-value service models—including AI-based performance optimization and life prediction solutions—into our proprietary product lines. **Industrial and Economic Impact** 1) Import Substitution and Cost Reduction ● By replacing ultra-precision stage systems that were entirely dependent on foreign sources (e.g., Germany, Japan) with domestic technology, we will drastically reduce manufacturing costs and contribute to the stability of the domestic semiconductor and display supply chains. 2) Elevation of National Technological Standing ● Joint research with advanced overseas institutions will elevate the technological level of Korean SMEs to a global standard and open new channels for overseas markets through reverse exports. 3) Enhanced Production Efficiency and Sustainability ● This technology will contribute to higher production efficiency by significantly reducing defect rates in next-generation display processes

		and support eco-friendly manufacturing by reducing driving energy through optimized control algorithms.

6. References

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